BUTTERFLYDREAMING · TECHNICAL NOTE

Grammars that draw and play

How a few lines of text become a figure, a sculpture, or a melody

Three of ButterflyDreaming's modules — a two-dimensional drawing, a three-dimensional one, and a piece of generative music — look unrelated on screen. They are the same idea three times over. Each is a short piece of text that grows into a much longer one, and is then read aloud by something that knows what the letters mean. This note explains the mechanism and, more usefully, explains where the line falls between the part that is written and the part that is adjusted by hand.

1. An L-system: a grammar that grows

An L-system is a rewriting rule. You begin with a short string, the *axiom*, and a set of *productions* of the form "wherever you see this symbol, put these symbols instead". Then you apply all the productions at once, and again, and again. Nothing is random and nothing is drawn yet; this stage is pure text.

Here is one actually stored in ButterflyDreaming, exactly as it appears:

```
axiom:  F+F+F+F+F+F+F+F
F:      F+F−F−F+F+F+F−F
```

The axiom is eight `F`s with turns between them. The single production says every `F` becomes fifteen symbols. So:

```
depth 0    F+F+F+F+F+F+F+F                    8 F
depth 1    each F replaced                   64 F
depth 2    again                            512 F   ← 1,023 symbols in total
```

Those numbers are not illustrative: that score is stored with a depth of 2, and the string it produces contains 1,023 symbols of which 512 are `F`. Two lines of text, five hundred strokes. This is the whole appeal — the growth is *steep*, so a human-sized input yields a figure no human would draw by hand, and yet it is entirely determined. The same two lines always give the same 1,023 symbols.

Symbols that draw nothing

Productions may use symbols that have no graphical meaning at all. The music module's default grammar is a Hilbert curve:

```
axiom:  X
X:      FYFX+F+YFXFY−F−XFYFX
Y:      FXFY−F−XFYFX+F+YFXFY
```

`X` and `Y` never draw. They exist only to be rewritten — carriers that steer how the `F`s and turns arrange themselves, and then vanish from the reading. This is worth knowing because it explains why the

alphabet of the grammar and the alphabet of the renderer are *different lists*, which is the subject of the next section.

2. The renderer: a string means nothing until something walks it

A thousand characters of `F`s and plus signs is not a picture. It becomes one only when a second piece of machinery reads it symbol by symbol and does something physical with each. The traditional device is a *turtle*: an imaginary pen with a position and a heading, which obeys instructions.

ButterflyDreaming's turtle understands exactly seven symbols:

SYMBOL	WHAT THE TURTLE DOES
F	**Draw** forward one step, and move there
f	**Move** forward one step without drawing — pen up
+	Turn anticlockwise by the current angle
−	Turn clockwise by the current angle
[	**Push** the present position and heading onto a stack
]	**Pop** — return to the last pushed position and heading
|	Turn 180°

Any other character is *ignored*. That single sentence is what lets `X` and `Y` exist. It also means a mistyped grammar fails quietly rather than loudly: a rule for a symbol the turtle does not know will rewrite happily and draw nothing.

The bracket pair is what separates a line from a tree. Without `[` and `]` the turtle traces a single unbroken path; with them it can set off down a branch, draw it, and return to the fork to continue. None of ButterflyDreaming's present figures use them, which is a choice rather than a limit — kolam are continuous line drawings by tradition.

> **One departure worth naming.** In a textbook turtle, `F` draws a straight line, and an L-system therefore produces polygons. ButterflyDreaming draws a *quadratic curve* instead, placing its control point perpendicular to the heading at three tenths of the step length. The grammar is unchanged; only the reading of `F` differs. That one substitution is most of why these figures look like kolam — looping and woven — rather than like a Koch snowflake.

3. The separation that makes it workable

Notice what the grammar does *not* say. `+` means "turn", not "turn ninety degrees". `F` means "draw a step", not "draw 299 pixels". The grammar fixes the **sequence** of operations and nothing else; the **magnitudes** come from elsewhere.

This is the architectural decision the whole system rests on, and its consequence is immediate: the same string can be redrawn at a different angle and become an entirely different figure. A turn of 90° closes an eight-step axiom into a square ring; 90°26' leaves it very slightly open, so each repeat lands rotated a little from the last and the figure precesses into a rosette. Not one character of the grammar changed.

Hence the parameters are expressed in minutes of arc as well as degrees. Whole degrees proved too coarse a control: the interesting figures live in the near-misses, where a shape almost closes.

4. Directives and sliders: two faces of one value

Every parameter is a line of text in the node itself. A directive is a line beginning `%%bd_`, and the drawing above is stored like this:

```
%%bd_module bd_V_Kolam
%%bd_p_symmetry 8          how many copies about the centre
%%bd_p_depth 2             how many rewriting passes
%%bd_p_step 299            how far F travels
%%bd_p_angle 90            how much + and − turn ...
%%bd_p_angle_minutes 26    ... to the minute of arc
%%bd_p_angle_drift 4       a slow animation of that angle
%%bd_p_colour_speed 4      how fast hue follows heading
%%bd_p_weight 0.5          stroke width
%%bd_p_opacity 0.76
%%bd_stroke angle          colour BY heading, rather than one colour
%%bd_background #0a0a0f
%%bd_score [
axiom: F+F+F+F+F+F+F+F
F: F+F−F−F+F+F+F−F
%%bd_]
```

The module also presents these as sliders and menus. The two are not parallel mechanisms: **the sliders are an editor for the text.** Moving one rewrites the corresponding `%%bd_p_` line in the script and re-renders from the rewritten script. There is no separate state of the world that the text might disagree with — which is why a figure arrived at by twenty minutes of dragging can be kept simply by saving the node.

Two details of that round trip show the care it needed:

- **A directive the controls do not manage is left exactly where it is.** Older scripts carry parameters that have since been renamed or withdrawn. The rewriter ignores them rather than deleting them, on the principle that removing an author's line is not a value update's business.
- **The prefix matters.** Parameters are `%%bd_p_name`; matching the bare name instead inserts a second line, and the parser takes the last one. A whole class of "the slider does nothing" reports came from exactly that.

The practical division, when something looks wrong: **if the topology is wrong it is the grammar; if the shape is right but the size, sharpness or colour is wrong it is a directive.**

5. Three readings of one grammar

The three modules share the grammar and differ in what the turtle's walk is taken to *mean*.

Kolam 2D — ink on a plane

The turtle walks a flat canvas and leaves curves. One extra stage sits outside the turtle entirely: `%%bd_p_symmetry 8` draws the finished path eight times about the centre. With 512 strokes and eightfold symmetry, roughly four thousand curves reach the screen. Hue can be tied to heading rather than fixed, so the direction of travel becomes visible as colour.

Kolam 3D — the same walk, lifted out of the plane

The grammar is *identical*, and so are the seven symbols. One thing is added: every turn now applies a **pitch** as well as a yaw, so `+` rotates in two planes at once and the path leaves the flat. `F` still draws a quadratic curve, now in three dimensions and tessellated into short segments so it can be given to a graphics card.

The clearest way to see that this is the same system: **set pitch to zero and the camera to directly overhead, and the 3D module reproduces the 2D figure.** Pitch is the single parameter that makes it a sculpture rather than a drawing, and the camera parameters — elevation, distance, and their drift rates — are a viewpoint on the result, not part of it.

Fractal Music — the same walk, read as a melody

Here the turtle still walks a flat grid, but nothing is drawn. The path itself is read as music:

- **Height becomes pitch.** A vertical position of zero is the root note low in the range; each unit upward is one degree up the chosen scale — minor, major, minor pentatonic, or blues. The melody is literally the shape of the curve.
- **Horizontal runs become notes.** A segment travelling sideways sounds; consecutive sideways segments at the same height become one longer note. The vertical moves are how the melody changes pitch, not events in themselves.

The default grammar is the Hilbert curve shown earlier — a space-filling curve, which is a fortunate choice for music because it wanders through every height in its range without ever leaping.

This module differs in one engineering respect worth recording. The other two hand their grammar to a small library which does the rewriting and returns the finished string. At the depths this one uses, that string is far too large to hold in memory, so it expands and walks *lazily* — depth first, consuming symbols as it goes, never materialising the whole. A stepper control then lets the listener start thousands of symbols into the walk, so a long piece can be entered in the middle.

	KOLAM 2D	KOLAM 3D	FRACTAL MUSIC
Grammar	shared	shared	shared
Symbols	the seven	the seven	`F + −` only
`F` **means**	a curve on a plane	a curve in space	a step of time
Turns act in	yaw	yaw and pitch	yaw
Height means	position	position	**pitch**
Rewriting by	library	library	lazy walk
Outer stage	symmetry copies	camera	scale, tempo, effects

6. Why it is all text

A figure in ButterflyDreaming is not a file. It is a passage of text in a node, and that has consequences the system is built around.

- **It is small enough to share.** The two lines of grammar and a dozen directives describe a four-thousand-stroke drawing. They travel in a message, a link, or a conversation.
- **It is editable by a reader.** Changing `angle` by a minute of arc is a plausible act for someone who is not a programmer, and the result is immediate.
- **It is reproducible — conditionally.** The same text gives the same figure, *provided* the machinery that reads it does not change underneath. That is not automatic. The rewriting library was for a time fetched unpinned from a public network, which meant a new release elsewhere could have altered stored figures with no change made here. It is now a fixed copy held alongside the code. When a figure *is* the artwork, the version of the thing that expands its grammar is part of the artwork.

> **The shape of the whole thing.** A grammar supplies order; a renderer supplies meaning; directives supply magnitude; and an outer stage — symmetry, camera, or scale and tempo — supplies presentation. Those four are genuinely separable, which is why the same two lines of text can be a drawing, a sculpture, and a tune, and why a slider can be handed to someone with no interest in grammars at all.

ButterflyDreaming · `butterflydreaming.net` · L-system rewriting by `lindenmayer` 1.5.4 (MIT, Tom Brewe); turtle, renderers and directive system are ButterflyDreaming's own. Figures, scores and parameter values quoted in this note are taken from the live corpus.